

Changes in the U.S. Wage Distribution, 2007–2024: A Propensity-Score Reweighting Analysis

Xiangzhi Zhao

October 7, 2026

1 Introduction

The U.S. wage distribution can change for at least two conceptually distinct reasons. First, the composition of the labor force may change. Workers in different years may differ in education, experience, demographic characteristics, union status, occupation, industry, and other attributes associated with wages. Second, the wage structure itself may change: workers with similar observable characteristics may receive different wages at different points in time. Distinguishing these two sources of change is important for understanding changes in wage inequality and the broader evolution of the labor market.

This paper examines changes in the U.S. real hourly wage distribution between 2007 and 2024 using data from the Current Population Survey (CPS). Following the propensity-score reweighting approach of DiNardo, Fortin, and Lemieux (1996), the analysis treats 2007 as the reference year and constructs a counterfactual distribution by reweighting workers observed in 2024 so that their observable characteristics resemble those of workers in 2007.

The resulting counterfactual asks what the 2024 wage distribution would have looked like if workers had faced the 2024 wage structure but had the observable composition of the 2007 workforce. This provides a way to assess how much of the observed change in wages is associated with changes in workforce composition rather than with the remaining changes in the wage distribution.

The descriptive evidence shows a substantial upward shift in real wages between 2007 and 2024. At the same time, the observable composition of the workforce changed substantially, particularly with respect to education, racial and ethnic composition, metropolitan residence, and union status.

The propensity-score results indicate that these compositional changes are economically important but do not account for the entire shift in the wage distribution. Using the CPS analysis weights, mean real hourly wages increased from approximately \$29.50 in 2007 to \$37.75 in 2024. At the mean, approximately 40.8 percent of this increase is associated with changes in observable workforce composition. The importance of composition varies considerably across the distribution, with larger effects around the middle and upper portions of the wage distribution than at the bottom.

2 Data and Analytic Sample

The analysis uses the 2007 and 2024 CPS Merged Outgoing Rotation Group samples obtained through IPUMS CPS. The data cover all twelve months of each year and are restricted to individuals eligible for the CPS earner sample. The CPS MORG is particularly useful for this exercise because it contains information on earnings and hours together with demographic, educational, geographic, industry, occupation, and union characteristics.

The analytic sample is restricted to wage and salary workers between ages 16 and 65, excluding self-employed workers and unpaid family workers. Usual weekly hours are constructed primarily from the basic monthly usual-hours measure. When this variable indicates that hours vary, the earner-study hours measure is used when a valid value is available. Observations with unresolved or invalid hours are removed. Observations whose earnings were allocated by the Census Bureau are also excluded.

Hourly wages are constructed using the reported hourly wage for hourly paid workers and usual weekly earnings divided by usual weekly hours for other workers. The 2007 wage measure is converted to 2024 dollars. Observations with real hourly wages below \$2 or above \$300 are excluded. The analysis weight is constructed by multiplying the CPS earner-study weight by usual weekly hours.

The worker characteristics used in the analysis include years of schooling, potential labor-market experience, sex, race and Hispanic ethnicity, marital status, part-time status, metropolitan residence, union status, Census region, industry, and occupation. Potential experience is constructed from age and schooling. Industry and occupation categories are allowed to be year-specific in the wage regressions because the underlying Census classification systems differ between 2007 and 2024.

Observations with missing values for variables required by the analysis are removed. The resulting regression sample contains 188,111 observations, of which 112,428 are from 2007 and 75,683 are from 2024.

Table 1: Descriptive Statistics by Year

Variable	2007		2024	
	Mean	SD	Mean	SD
Real hourly wage	28.367	18.603	35.942	34.274
Years of schooling	13.615	2.630	14.331	2.695
Potential experience	20.859	12.619	21.109	12.951
Female	0.500	0.500	0.495	0.500
Black	0.081	0.273	0.085	0.279
Asian	0.038	0.190	0.067	0.250
Hispanic	0.121	0.326	0.169	0.375
Married	0.566	0.496	0.533	0.499
Part-time	0.183	0.387	0.165	0.371
Metropolitan	0.794	0.404	0.834	0.372
Unionized	0.140	0.347	0.114	0.318
Observations	112,428		75,683	

Notes: The table reports unweighted descriptive statistics for the final analytic sample. Wage values are expressed in 2024 dollars. Binary-variable means can be interpreted as sample proportions. The propensity-score reweighting and counterfactual estimates reported below use CPS analysis weights.

Table 1 reveals meaningful compositional changes. Average schooling increased by approximately 0.72 years, from 13.61 to 14.33 years. The Asian share increased from 3.8 percent to 6.7 percent, and the Hispanic share increased from 12.1 percent to 16.9 percent. Metropolitan residence increased from 79.4 percent to 83.4 percent.

In contrast, unionization declined from 14.0 percent to 11.4 percent, and the married and part-time shares also declined. These changes provide a substantive motivation for reweighting: a simple comparison of the two wage distributions combines changes in worker composition with changes in the prices associated with worker characteristics.

3 Methods

The empirical problem is that the observed wage distributions in 2007 and 2024 reflect both differences in the types of workers observed in each year and differences in how the labor market rewards those characteristics. A direct comparison therefore does not by itself identify how much of the change is associated with workforce composition.

Following DiNardo, Fortin, and Lemieux (1996), I use propensity-score reweighting to construct a counterfactual 2024 distribution. Define

$$D_i = \begin{cases} 1 & \text{if individual } i \text{ is observed in 2007,} \\ 0 & \text{if individual } i \text{ is observed in 2024.} \end{cases}$$

I estimate a logistic regression of year status on observed worker characteristics:

$$p(X_i) = P(D_i = 1 \mid X_i) = \Lambda(X_i' \beta), \quad (1)$$

where X_i includes age, age squared, years of schooling, sex, race and Hispanic ethnicity, marital status, part-time status, metropolitan residence, union status, and Census region. The final propensity-score specification excludes detailed industry and occupation indicators. Because the underlying detailed classifications differ across the two years, entering these categories directly generates severe prediction problems and undermines overlap between the samples.

For observations from 2024, the estimated odds of belonging to the 2007 sample provide the basis for the DiNardo–Fortin–Lemieux reweighting factor:

$$\omega_i = \frac{\hat{p}(X_i)}{1 - \hat{p}(X_i)}. \quad (2)$$

The reweighting factor is combined with the original CPS analysis weight. Intuitively, a 2024 worker with characteristics that were relatively common in 2007 receives greater weight, whereas a worker whose characteristics are disproportionately associated with 2024 receives less weight. The resulting weighted 2024 sample is intended to reproduce the observable composition of the 2007 workforce while retaining wages observed under the 2024 wage structure.

The validity of the reweighting exercise depends on sufficient overlap in observed characteristics between the two years and on the ability of the weights to balance the included covariates. I therefore examine the propensity-score distributions across years and evaluate covariate balance using standardized mean differences before and after reweighting. I also inspect the distribution of the reweighting factors for potentially influential observations.

4 Results

4.1 Changes in the Wage Distribution

The real hourly wage distribution shifted upward substantially between 2007 and 2024. In the descriptive sample, mean real hourly wages increased from \$28.37 to \$35.94, while the median increased from \$22.69 to \$26.00.

The increase occurred throughout the distribution rather than only at the mean. The 10th percentile increased from \$11.65 to \$15.00, the 25th percentile from \$15.13 to \$18.00, the 75th percentile from \$35.55 to \$40.85, and the 90th percentile from \$52.75 to \$65.00.

Table 2: Real Hourly Wage Distribution, 2007 and 2024

Statistic	2007	2024
Mean real hourly wage	28.37	35.94
Standard deviation	18.60	34.27
10th percentile	11.65	15.00
25th percentile	15.13	18.00
Median	22.69	26.00
75th percentile	35.55	40.85
90th percentile	52.75	65.00
Mean log wage	3.167	3.345
SD log wage	0.591	0.633

The increase was particularly pronounced toward the upper portion of the distribution. The P90–P10 gap increased from approximately \$41.10 in 2007 to \$50.00 in 2024. Consistent with this pattern, the standard deviation of log wages increased from 0.591 to 0.633. Thus, the data show both an upward movement in real hourly wages and evidence of increased dispersion.

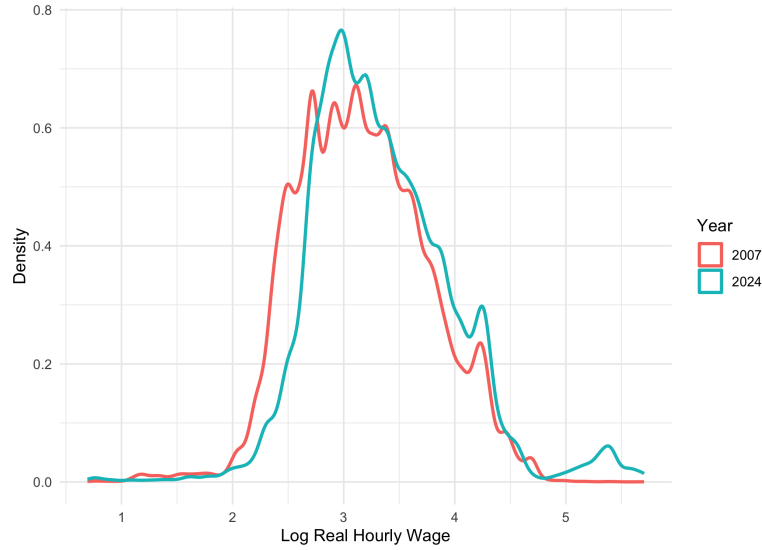


Figure 1: Density of Log Real Hourly Wages, 2007 and 2024

Figure 1 compares the distributions of log real hourly wages. The 2024 distribution is shifted toward higher log wages, although the two distributions retain substantial overlap.

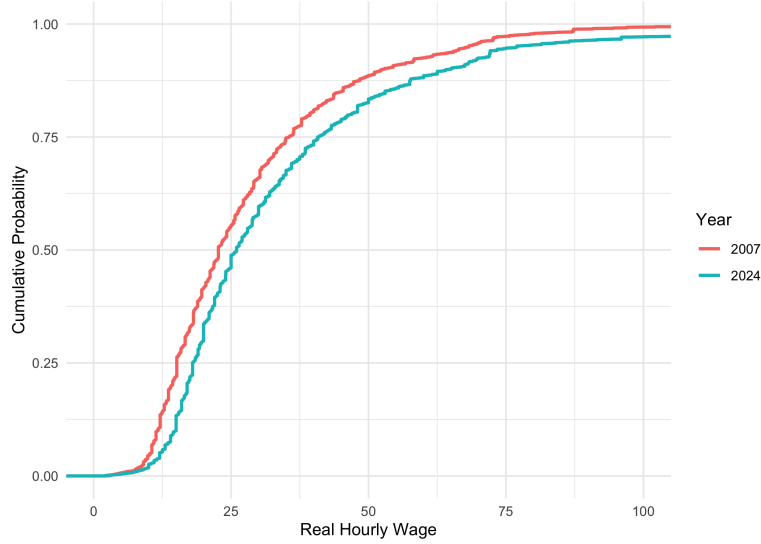


Figure 2: Empirical CDF of Real Hourly Wages, 2007 and 2024

Figure 2 reinforces this result. Over most of the relevant wage range, the 2024 CDF lies below and to the right of the 2007 CDF, indicating that a smaller fraction of 2024 workers earn below a given real hourly wage threshold. The descriptive statistics and graphical evidence therefore consistently indicate an upward shift in the wage distribution.

4.2 Propensity Score Diagnostics

The propensity-score model estimates the probability that an individual belongs to the 2007 rather than the 2024 sample as a function of observed worker characteristics. The estimated propensity scores exhibit substantial overlap across the two years. The mean propensity score is 0.512 for workers observed in 2007 and 0.470 for workers observed in 2024. The corresponding 1st–99th percentile ranges are approximately 0.269–0.708 and 0.234–0.684, respectively. The observed ranges therefore indicate considerable common support.

More importantly, reweighting substantially improves covariate balance. Table 3 reports standardized mean differences before and after reweighting. Before reweighting, the largest imbalance occurs in years of schooling, with an absolute standardized mean difference of 0.282. Differences are also visible for Asian and Hispanic ethnicity and metropolitan residence. After reweighting, these differences become very small. The maximum absolute standardized mean difference falls from 0.282 before reweighting to only 0.010 afterward.

Table 3: Covariate Balance Before and After Reweighting

Variable	SMD Before	SMD After
Age	0.077	-0.001
Years of schooling	0.282	-0.006
Female	0.024	-0.010
Black	0.059	-0.003
Asian	0.133	0.004
Hispanic	0.115	-0.005
Married	-0.082	-0.008
Part-time	-0.021	-0.001
Metropolitan	0.108	-0.001
Unionized	-0.068	0.003
Maximum absolute SMD	0.282	0.010

Notes: Standardized mean differences compare the original 2024 sample with the 2007 reference sample before reweighting and the counterfactually weighted 2024 sample with the 2007 sample after reweighting.

The improvement in balance is also apparent from the weighted covariate means. Average schooling is 13.65 years in the weighted 2007 sample and 14.42 years in the original weighted 2024 sample. After reweighting the 2024 sample, average schooling falls to 13.64 years. Similarly, the Asian share changes from approximately 7.0 percent in the original 2024 sample to 4.4 percent after reweighting, close to the 4.3 percent observed in the weighted 2007 sample. Metropolitan residence changes from approximately 88.9 percent to 85.0 percent after reweighting, essentially matching the 85.0 percent rate in the weighted 2007 sample.

The reweighting factors for 2024 observations are also reasonably bounded. Their median is approximately 0.89, the 95th percentile is 1.73, the 99th percentile is 2.16, and the maximum is approximately 8.31. Together with the propensity-score overlap and the substantial improvement in standardized mean differences, these diagnostics suggest that the reweighting procedure successfully reproduces the included observable characteristics of the 2007 reference population.

4.3 Counterfactual Wage Distribution

The central decomposition compares three distributions: the observed 2007 wage distribution, the observed 2024 wage distribution, and a counterfactual 2024 distribution obtained by reweighting 2024 workers to reproduce the observable composition of the 2007 workforce. The counterfactual therefore retains wages observed under the 2024 wage structure while holding the included observable workforce characteristics approximately at their 2007 distribution.

At the mean, the weighted real hourly wage increased from \$29.50 in 2007 to \$37.75 in 2024, an

observed increase of approximately \$8.24 per hour. The counterfactual 2024 mean is \$34.39. Thus, approximately \$3.36 of the observed increase is associated with changes in observable workforce composition, while the remaining \$4.88 persists after reweighting.

Expressed as shares of the total increase, observable workforce composition accounts for approximately 40.8 percent of the increase in the mean, while approximately 59.2 percent remains after holding the included observable characteristics approximately fixed.

Table 4: Decomposition of the Change in Mean Real Hourly Wages

Statistic	Value
Mean wage, 2007	\$29.50
Mean wage, 2024	\$37.75
Counterfactual 2024 mean	\$34.39
Observed change	\$8.24
Composition component	\$3.36
Remaining component	\$4.88
Composition share	40.8%
Remaining share	59.2%

The average decomposition masks substantial heterogeneity across the wage distribution. Table 5 reports actual and counterfactual wages at selected percentiles.

Table 5: Actual and Counterfactual Real Hourly Wage Quantiles

Percentile	2007	2024	Counterfactual 2024	Composition Share
P10	12.10	15.00	15.00	0.0%
P25	15.89	19.10	18.25	26.4%
P50	24.21	27.00	25.00	71.6%
P75	36.78	43.25	38.55	72.7%
P90	55.73	67.50	60.00	63.7%

Notes: The counterfactual distribution reweights workers observed in 2024 to reproduce the included observable characteristics of the 2007 workforce. The composition share equals the difference between the actual and counterfactual 2024 statistic divided by the total difference between the actual 2024 and 2007 statistics.

The decomposition varies substantially across the wage distribution. At the 10th percentile, the actual and counterfactual 2024 wages are both \$15.00. Reweighting therefore produces essentially no composition component at this point in the distribution. At the 25th percentile, observable composition accounts for approximately 26.4 percent of the observed increase.

The role of observable composition becomes substantially larger around the middle and upper

portions of the distribution. At the median, the actual 2024 wage is \$27.00, compared with a counterfactual value of \$25.00 and a 2007 value of \$24.21. Observable composition therefore accounts for approximately 71.6 percent of the observed increase at the median. The corresponding composition share is approximately 72.7 percent at the 75th percentile and 63.7 percent at the 90th percentile.

These results indicate that compositional changes played very different roles at different points of the wage distribution. The increase near the bottom of the distribution is largely preserved when the 2024 sample is reweighted to the 2007 observable composition. In contrast, a considerably larger portion of the increase around the middle and upper portions of the distribution is associated with changes in the observable characteristics of workers.



Figure 3: Actual and Counterfactual Real Hourly Wage Distributions

Figure 3 provides a graphical representation of the same decomposition. The counterfactual 2024 CDF generally lies between the observed 2007 and observed 2024 distributions. Reweighting therefore shifts the 2024 distribution toward the 2007 distribution without eliminating the entire difference.

The movement of the counterfactual distribution toward the 2007 distribution is particularly visible through the middle and upper portions of the distribution. This graphical pattern is consistent with the quantile decomposition, which indicates substantially larger composition shares at the median, 75th percentile, and 90th percentile than at the bottom of the wage distribution.

5 Discussion and Conclusion

This paper examines changes in the U.S. real hourly wage distribution between 2007 and 2024 and asks how much of those changes are associated with shifts in observable workforce composition. The descriptive evidence shows a substantial upward movement in real wages over this period together with economically meaningful changes in the characteristics of the workforce.

Propensity-score reweighting successfully removes most of the measured compositional differences between the two samples. The maximum absolute standardized mean difference across the included covariates falls from 0.282 before reweighting to 0.010 afterward. This substantial improvement in balance indicates that the reweighted 2024 sample closely reproduces the included observable characteristics of the 2007 workforce.

The mean decomposition indicates that observable workforce composition accounts for an economically important, but incomplete, portion of wage growth. Weighted mean real hourly wages increased by approximately \$8.24 between 2007 and 2024. Reweighting reduces the 2024 mean from \$37.75 to \$34.39, implying a composition component of approximately \$3.36, or 40.8 percent of the observed increase. The remaining \$4.88, or 59.2 percent, persists after holding the included observable characteristics approximately fixed.

The distributional results reveal substantial heterogeneity behind this average effect. Observable composition explains essentially none of the increase at the 10th percentile and approximately one quarter of the increase at the 25th percentile. In contrast, it accounts for roughly 72 percent of the increase at both the median and the 75th percentile and approximately 64 percent at the 90th percentile. Changes in workforce composition therefore appear to have played a considerably larger role in the middle and upper portions of the wage distribution than at the bottom.

These results should not be interpreted as a causal decomposition of all forces affecting wages. The reweighting procedure balances the samples only on characteristics included in the propensity-score model. Unobserved changes in worker skills, job characteristics, labor demand, technology, labor-market institutions, and other economic conditions may contribute to the remaining component. The counterfactual exercise also relies on sufficient common support between the two samples.

A further limitation is that detailed industry and occupation classifications are not incorporated directly into the final propensity-score specification. The coding systems differ across the two years, and using the detailed categories to predict year status generates severe separation and overlap problems. The counterfactual should therefore be interpreted as holding fixed the included observable demographic and employment characteristics rather than every potentially relevant dimension of workforce composition.

Subject to these qualifications, the results show that the change in the U.S. wage distribution between 2007 and 2024 cannot be characterized solely as a consequence of either workforce composition or changes in the wage structure. Observable compositional change accounts for a substantial fraction of the increase in wages, particularly in the middle and upper portions of the distribution, while a sizable remaining shift persists even after those observable characteristics are held approximately fixed.

References

- [1] DiNardo, John, Nicole M. Fortin, and Thomas Lemieux. “Labor Market Institutions and the Distribution of Wages, 1973–1992: A Semiparametric Approach.” *Econometrica*, 64(5), 1996, pp. 1001–1044.
- [2] Flood, Sarah, Miriam King, Renae Rodgers, Steven Ruggles, J. Robert Warren, and Michael Westberry. *Integrated Public Use Microdata Series, Current Population Survey: Version 12.0*. Minneapolis, MN: IPUMS.